

Spatial Computing Opportunities in Biomedical Decision Support: The Atlas-EHR Vision

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We consider the problem of reducing the time that healthcare professionals need to understand the patient's medical history through the next generation of biomedical decision support. This problem is societally important because it has the potential to improve healthcare quality and patient outcomes. However, navigating electronic health records (EHR) is challenging due to high patient-doctor ratios, potentially long medical histories, urgency of treatment for some medical conditions, and patient variability. The current EHR systems provide only a longitudinal view of patient medical history, which is time-consuming to browse, and doctors often need to engage nurses, residents, and others for initial analysis. To overcome this limitation, we envision an alternative spatial representation of patient histories (e.g., electronic health records) and other biomedical data in the form of Atlas-EHR. Just like Google Maps, which allows a global, national, regional, and local view, Atlas-EHR can start with an overview of the patient's anatomy and history before drilling down to spatially anatomical subsystems, their individual components, or subcomponents. Atlas-EHR presents a compelling opportunity for spatial computing since healthcare is almost a fifth of the US economy. However, traditional spatial computing designed for geographic use cases (e.g., navigation, land survey, and mapping) faces many hurdles in the biomedical domain. This article presents several open research questions under this theme in five broad areas of spatial computing.

CCS Concepts: • **Computing methodologies** → **Spatial and physical reasoning**; **Temporal reasoning**
• **Cross-computing tools and techniques** → **Design**;

Additional Key Words and Phrases: Atlas-EHR, biomedical decision support, inner space, spatial computing, vision

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1 Introduction

Electronic health records (EHRs) and other biomedical data are critical sources of information for healthcare professionals, providing them with valuable information on patient health and medical history. However, the sheer amount of data can be time-consuming for healthcare professionals to navigate. Towards this goal, we envision the next generation of biomedical decision support, termed Atlas-EHR, which leverages advanced spatial computing, AI-driven predictive insights, and real-time data integration to revolutionize the way medical histories are accessed and interpreted. This system aims to enable healthcare providers and possibly other users to investigate the human body by overlaying medical records onto a dynamic 4-dimensional anatomical display, incorporating both spatial and temporal dimensions, and spanning various levels of detail from organ-level relationships to sub-cellular processes. By simplifying access to comprehensive spatially organized patient data, Atlas-EHR aims to reduce the time that healthcare professionals spend examining and understanding the medical history from the EHR data.

Societal Importance: This problem is societally important since it helps healthcare professionals make efficient and accurate informed decisions. This may ultimately improve the quality of care and patient outcomes. Also, by rethinking how medical professionals access and interpret patient medical histories, they may have more time to spend and interact with patients, resulting in improved healthcare standards.

Challenges: However, the problem is challenging due to several reasons. First, patient visits are short (e.g., around 15 minutes). In addition, a patient's medical history can be long spanning decades. Furthermore, many primary care doctors are responsible for hundreds and thousands of patients. Additionally, changes in medical guidelines and practices introduce additional complexities. Some medical conditions also require treatment urgently. Further, some medical decisions are sensitive to patient demographics and underlying medical conditions. Moreover, ensuring explainability, fairness, accountability, security, and trustworthiness in EHR systems is paramount challenge for the advancement of biomedical decision support. Explainability improves patient care by enabling healthcare providers to understand and trust the system's recommendations. Fairness is vital to mitigate biases that could influence treatment outcomes, especially considering the diverse interplay of spatial and location variability, demographics, genetics, and lifestyle. Accountability and security are fundamental for safeguarding sensitive patient information to maintain privacy and adhering to healthcare regulations. Trustworthy EHR systems that embody these attributes are indispensable for delivering informed and equitable biomedical decision support.

The other technical aspects include the standard spatial data models and query techniques, which are limited to capturing the complex, dynamic, and non-rigid processes in the human body. For example, modeling and querying blood flow requires knowledge of the continuous pulsatile characteristics of blood movement through flexible vessels that can deform in both shape and size. Conventional models such as R-trees or Voronoi diagrams are not designed to handle the dynamic, multi-scale, and heterogeneous characteristics of biological systems. Another challenge is spatial variability, which refers to the location-dependent variation observed in patterns across the space (i.e., the human body). For example, consider the varying density and distribution of capillaries in different tissues. In the lungs, capillaries are densely packed to facilitate gas exchange, while in muscle tissue, their densities vary with activity levels and metabolic demands. This variability

requires models that dynamically adjust to various spatial patterns and scales to provide accurate and context-specific insights. Traditional navigation algorithms, such as Dijkstra's or A*, do not apply directly because there is no fixed coordinate system and the variability in human anatomy. Developing a navigation system for medical procedures requires a context-aware spatial indexing method that adapts to patient-specific anatomical variations. The article details more technical challenges within five broad areas of spatial computing.

Related Work: Popular EHR systems (e.g., Epic Systems [27]) provide a longitudinal view of a patient's history, which makes it time-consuming for medical professionals to navigate through (See Figure 10 for illustration). This representation does not allow medical professionals to explore the interdependencies of EHRs data generated through different sensing modalities over various timescales spanning decades from the whole body to their components or sub-components. Early spatial vision approach includes inner space [114]. The article discussed harnessing traditional medical imaging (e.g., X-ray and MRI) data and the concept of spatial networks to build a spatiotemporal database. Their spatial framework was supposed to answer long-term questions about how a disease would progress or the comparative effectiveness of a therapeutic intervention. Challenges they identified at the time included the lack of a spatial reference frame for the human body, how to determine the location in the body, routing in a continuous space, observing change across snapshots, and scalability. More recent work [159] explored the human brain as a spatiotemporal object. It highlighted the challenges of GIS-inspired methods (e.g., spatial aggregation for brain parcellations) to study the brain structure and signals communicated with it, which dynamically change depending on the mental task or the function performed by the person. Other literature work on the **human cell atlas (HCA)** [19, 60, 61, 125, 129, 132, 151, 168] and the **human biomolecular atlas program (HuBMAP)** [51, 57, 73, 82, 83] along with the early emergence of digitized human anatomies [14, 93, 112] to form a reference map as a basis for understanding human health to improve diagnosis, prognosis, monitoring, and treatment of disease. However, these studies are either limited to specific types of EHR data and lack a comprehensive perspective regarding the EHR data across the human body as a system of systems [1, 72] through the lens of spatial computing research. A more detailed description of the HCA initiative is provided in Section 3.

Our Atlas-EHR Vision: We envision an alternative unified spatial framework for EHR and other biomedical data, which we call Atlas-EHR. These spatial representations, akin to digital maps like Google Maps, allow users to navigate from a global view of the patient's anatomy and history to more detailed views of anatomical sub-systems and their individual components or sub-components. Figure 1 illustrates a perspective of Atlas-EHR, where patient medical records are spatially organized according to their anatomy, integrating various structures from organs to tissues and cells. For example, it shows how medical imaging (e.g., heart electrocardiogram time series data) and clinical notes (e.g., medical prescriptions) can be spatially grouped based on relevant organs, such as the heart.

Furthermore, Atlas-EHR extends beyond cartography and geo-visualization, encompassing areas such as spatial data science [7, 138, 139, 142] and **spatial database management systems (SDMS)** [38, 39]. It leverages recent advancements in computer infrastructures (e.g., graphical processing units (GPUs) [107, 118]), cloud computing [55, 104], edge computing [20, 25, 144], and developments in artificial intelligence and machine learning [63, 84, 115, 124, 130]. Moreover, Atlas-EHR leverages immersive technologies such as **augmented reality (AR)** and **virtual reality (VR)**. The applications of immersive technologies may be varied, ranging from educational uses to surgical simulations [34, 102, 103, 160]. An illustrative example is a real-time AR captioning tool with object detection capabilities, which can elucidate the steps of a medical procedure and emphasize different anatomical components, like substructures within the human body. For instance,

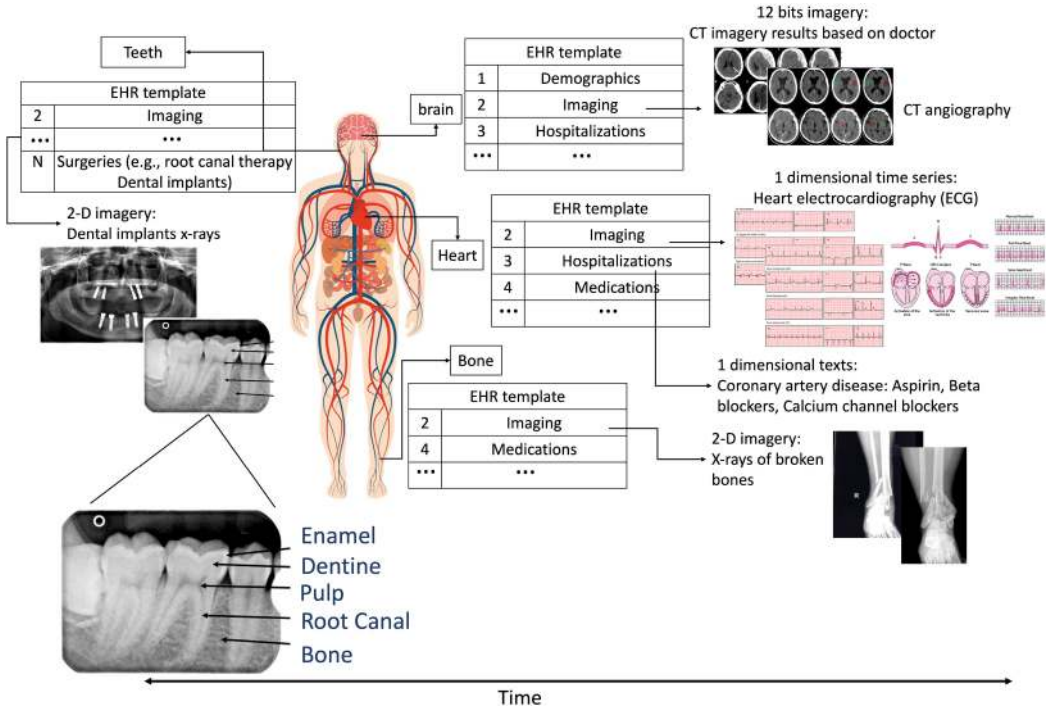


Fig. 1. A spatially-organized map of EHRs.

consider an X-ray of teeth shown on the left side of Figure 1: an AR system could highlight various parts of the dental structure and, based on the severity of tooth decay, might suggest interventions such as fillings or root canal treatments. Furthermore, integration with IoT devices [2, 87, 120, 127], including smartwatches and body-worn sensors, further enhances Atlas-EHR’s capabilities.

We envision healthcare providers being able to perceive not only the “what” but also the “where” and “how” of patient data, elements often overlooked in conventional methods of data presentation. Spatial computing has the potential to transform EHRs into dynamic, interactive maps of human health. Atlas-EHR not only represents the current state of a patient’s health but also provides predictive insights based on observed spatial and temporal patterns. This endeavor has the potential to bridge the gap between the spatial computing research community and the biomedical domain, enabling medical professionals to easily navigate EHRs and other biomedical data using a spatial approach. In the following sections, we highlight some of the recent breakthroughs in biomedical decision support that were not possible a decade ago and explain how these advancements can be integrated into our Atlas-EHR vision through the lens of spatial computing to improve healthcare quality. Atlas-EHR poses many challenges and research opportunities for current spatial computing, which is traditionally designed for geographic use cases (e.g., navigation, land survey, and mapping).

An Illustrative Example: Consider a senior patient with a history of hypertension and diabetes. He wears a smart wristband that continuously monitors his heart rate, blood pressure, and blood glucose levels. Additionally, he undergoes regular MRI scans and lab blood tests to monitor his cardiovascular health. Health professionals currently review the protensive segments of his EHR, including medical imaging, lab results, and clinical notes to comprehend his current conditions. This process involves analyzing EHR data from the same or different organs, which may or

may not be related, and cross-referencing these disparate data points. This is a time-consuming process and is prone to errors due to the fragmented nature of the data.

In Atlas-EHR, the patient's wearable device data, MRI images, and lab results are sent in real-time to be stored in a spatially-organized database. The system uses spatial indexing based on relevant organs (such as the brain, heart, and kidney) and further organizes data by anatomical regions (e.g., cardiovascular systems) and temporal sequences (e.g., chronological health events) to ensure quick and efficient access to specific data points. SDMS are designed to manage extensive and dynamic multi-dimensional biomedical data efficiently to safeguard its security, integrity, and ease of access. The spatial indexing capabilities of SDMS enable efficient storage and retrieval of data organized by anatomical regions and temporal sequences. Healthcare professionals, for example, cardiologists, use Atlas-EHR to query data related to the patient's cardiovascular systems. The system quickly retrieves recent MRI scans and displays trends in the patient's heart rate and blood pressure from a given temporal window which allows doctors to focus on the most relevant information without having to manually sift through disparate records.

The spatial pattern mining algorithms, including colocation pattern mining [140], can help identify significant patterns in heart rate variability and blood pressure readings that could be an indicator of potential heart failure. For example, if there is a colocation of high levels of some specific enzymes with abnormal electrical impulses, this can be noticed as a result of the onset of heart failure. An AI model predicts an increased risk of heart failure within the next three months and advises immediate intervention. Spatial context allows pinpointing where precisely the cardiovascular system problems occur, thereby making the intervention given very clear and actionable by the healthcare worker.

The interactive model of his cardiovascular system in Atlas-EHR highlights areas of concern with cautious and emergency colors. It includes MRI images and real-time data from wearable devices, providing a clear view of his heart health. The cardiologist can interact with the model, zooming in on specific areas to see detailed information. For example, she can overlay lab results and visualize how specific biomarkers are distributed in relation to the heart's anatomy and current function. Based on data interpretation and feedback, the process may loop back to data collection for additional data and model refinement. This continuous improvement cycle helps enhance the system's accuracy and effectiveness. For instance, after reviewing the predictions, the cardiologist may schedule additional tests for the patient, such as an echocardiogram or a stress test, and update his treatment plan based on these results. The new data from these tests are fed back into the Atlas-EHR system, further improving the accuracy of the AI models and predictive analytics.

This article makes the following contributions:

- It envisions a spatially-organized map of EHR data for reducing the time needed by healthcare to understand patient history via the next generation of biomedical decision support. Our vision is more comprehensive (e.g., EHR) than related works (e.g., inner space [114]).
- It challenges and opportunities for five broad areas of spatial computing research to explore the human body.

Scope: This is a vision article; hence prototyping, implementation, and experimental studies fall outside the scope of this article. A detailed description of spatial computing research in geographic space is also outside the scope of this article. Interested audiences can refer to available books and surveys [54, 138, 142, 164]. This article also does not address biological interpretation or mechanisms of domain-related applications in clinical trials. We use the term “inner space”¹ term

¹Innerspace [158] is a sci-fi movie directed by Spielberg that retells the story of *Fantastic Voyage* [70], in which a miniature submarine traverses the human body to examine the brain, blood arteries, eyes, and ears.

Table 1. Taxonomy of Topics in This Article

Spatial Computing Topic Study Area	Spatial Database Management	Spatial Pattern Mining (SPM)	Positioning, Navigation, Location-based services	Sensing	Cartography & Visualization
Geographic Space	OGC's Simple feature types [48], R-trees [49], long-visit duration query [135]	Ring-shaped hotspots [36], Local colocations [52, 53, 90], CNN [11], GWR [18], Physics-guided gap detection [136, 137], SVANN [59]	GPS, WiFi	RGB, Lidar, Multi-spectral, Hyper-spectral	GIS (e.g., ArcGIS)
Inner Space	Hadoop-GIS [3, 4], Spatial-hadoop [39]	CSCD [89], SRNet [88], Spatially Explainable Classifiers [41]	Talairach coordinate system [45]	X-rays, MRI, CT, MxIF [91]	IonPath [106]
Outer Space					

to refer to the human body and its inner parts as a fine biomedical scale for distinguishing from the geographic data used in traditional spatial computing (e.g., GPS and location-based services). Table 1 provides a taxonomy of the spatial computing methods discussed in this article.

Outline: The rest of this article is organized as follows: Section 2 provides an overview of recent innovations and outlines several challenges and opportunities in the field of spatial computing. Section 3 provides a brief background of spatial computing and an overview of emerging spatial datasets at the cellular level. Section 4 discusses research needs across five broad areas of spatial computing research in inner space. Section 5 briefly discusses the potential of Atlas-EHR and provides examples across a wider range of medical fields. Finally, Section 6 concludes the article and outlines directions for future research.

2 Challenges and Opportunities: Technological Evolution in EHR and Spatial Computing

The remarkable growth in EHRs coupled with recent advances in data handling by AI and spatial computing make this an ideal time to pursue an Atlas-EHR initiative. The **Patient Protection and Affordable Care Act (PPACA)** of 2010, commonly known as Obamacare, has promoted the continuous development of EHRs [56], aiming to reduce healthcare costs and improve the quality of healthcare and clinical research. This includes facilitating recruitment for clinical trials, easing burdensome data collection, and addressing uncertainty in generalizing results, as detailed in [28]. However, EHR systems still encounter significant limitations and challenges. Key issues include interoperability, privacy, security, data ownership, governance, data quality and validation, complete data capture, and heterogeneity between systems, among others [28, 66, 148]. In recent years, there have been efforts to consider the role of emerging, big, complex, and heterogeneous EHR data in healthcare and its applications [5, 10, 31, 80, 95, 163]. Yet, the full integration of this data across the spatial and temporal dimensions of the human body—an effort represented by Atlas-EHR—remains an emerging area of research and has not yet been fully realized.

For decades, designing pattern recognition and machine learning systems required deep engineering and domain-specific knowledge, particularly for creating effective feature extractors. These extractors convert raw data (e.g., image pixels), into feature vectors that the system's classifier then uses for pattern recognition in the data. The rise of deep learning techniques and the training of neural network architectures with billions, and more recently, trillions of parameters (e.g., Google Switch Transformers [43]), have significantly streamlined this intensive and computationally expensive process. This breakthrough has been made possible by the advent of fast graphics processing units (GPUs), along with high-performance computing centers, cloud computing, edge computing, and parallel computing techniques [32, 116, 156]. These GPUs, convenient to program, have enabled researchers to train networks 10 to 20 times faster [84]. The application of deep learning in medical imaging is a prime example of this evolution. Machine learning

models, such as convolutional neural networks (CNNs), have significantly improved the accuracy and speed of disease diagnosis from medical images [94, 143]. More recent techniques include the rise of graph convolutional neural networks (GCNNs) [162, 170], which are well-suited for unstructured data, such as graph-based data commonly found in molecular structures, chemical compounds, and spatial pathology datasets (e.g., cellular maps). These datasets are critical for drug discovery [24], protein-protein interaction networks [74], and gene-disease associations [123].

Deep Neural Networks: The success of CNNs in various applications, particularly in image recognition, is largely attributed to their ability to model spatial auto-correlation. CNNs prioritize spatially nearby samples over distant ones, capturing the inherent structure in images. This spatial awareness allows CNNs to outperform traditional fully-connected networks by a significant margin [79, 85]. However, a key challenge in handling spatial data, such as that found in EHRs, is spatial variability, that is the inherent heterogeneity and variations observed in spatial patterns, structures, properties, or arrangements across different locations [59]. The current deep learning literature [58] often adopts a spatial one-size-fits-all approach, training single **deep neural networks (DNN)** that do not account for spatial variability. Preliminary works have aimed to address this challenge by considering location-dependent weights, learning space partitionings of heterogeneous data, and, more recently, through the use of meta-learning [59, 92, 165]. However, the applicability of these methods to EHRs and healthcare applications, given the multi-modal and multi-scale nature of EHR data remains largely unexplored.

Foundation Models: More recently, the AI community has witnessed significant breakthroughs in foundation models [15], which can be fine-tuned for a broad range of tasks without needing task-specific data from scratch. Although a recent study [101] began to address the challenges associated with developing foundation models for geospatial AI, the potential of transferring these models from geospatial to biomedical scales remains largely unexplored. This gap offers a promising opportunity for the spatial computing community to innovate and broaden the application of foundation models to EHRs. Investigating additional data modalities beyond images and text, along with integrating the temporal aspects of EHRs, are crucial open research areas. An equally paramount challenge lies in ensuring spatial explainability [41], which means the ability to describe results using spatial concepts (e.g., touch and inside) and patterns (e.g., colocation). This is particularly relevant to oncologists seeking to advance new cancer therapies, as the spatial proximity between immune and cancer cells is crucial for evaluating the effectiveness of immune therapies. We further elaborate on this issue and explore related open research questions in Section 4.

Spatial Data Models: Recent innovations in SDMS, such as Spatial Hadoop [39], GeoMesa [46], and GeoSpark [167], have revolutionized data processing capabilities, making large-scale, complex spatial computations both feasible and efficient. These systems typically utilize established spatial data models, like **Open Geospatial Consortium's (OGC)** simple features (points, lines, and polygons), to store, query, and manipulate large spatial datasets. However, challenges arise when modeling the human body and its organs, which are non-rigid and dynamic exhibits significant variability in size, shape, texture, and function. How do we define boundaries (e.g., minimum orthogonal boundary rectangles (MOBRs)) for spatial indexing techniques (e.g., R, R+ trees) given the non-rigid and soft shapes of the human body and its organs? Unlike geographic space, the human body lacks a reference system analogous to latitude/longitude, which is a challenge for spatial computing. Traditional models like the **dimensionally extended 9-intersection model (DE-9IM)** offer a framework for understanding spatial interactions (e.g., touch and inside). Yet, adapting these to account for hierarchical relationships and the dynamic changes associated with growth or disease that are time-dependent and subject to real-time changes remains a key challenge. For instance, the movement and deformation of body parts, such as muscles and joints, alter

spatial relationships in intricate ways (e.g., parallel when the arm is extended and they form an acute angle upon bending). Additionally, fluid movement within body cavities, like blood in vessels or air in respiratory pathways, involves dynamic interactions that challenge static spatial modeling. Certain relationships, like bacteria within the gut or within a cell, are “inside” not just in terms of spatial positioning but also in terms of their functional roles and dependencies. Addressing these issues presents the spatial computing community with unique opportunities to explore.

3 Background and Emerging Spatial Data from Inner Space

3.1 Brief Overview of Spatial Computing

Spatial Computing [35, 139, 142] encompasses a diverse array of tools, frameworks, solutions, and technologies, collectively transforming our understanding and interaction with spatial data. This transformation extends beyond simple navigation; it influences how we perceive, communicate, and visualize relationships with various locations. A prime example is the Global Positioning System (GPS), a critical infrastructure for the world economy considering over 2 billion receivers in use for location and time services, with widespread applications from COVID-19 response to smart city development.

One of the main areas of spatial computing research is cartography and geo-visualization [29, 96, 97, 98]. This field extends beyond conventional data visualization by incorporating spatiotemporal elements and enables the representation and analysis of data to its physical or spatial location over time. Geographic information systems, such as ArcGIS [40] and QGIS [121], are widely used applications in diverse domains, such as public health, public safety, and agriculture. These systems offer a user-friendly interface to import data from third-party sources (e.g., the US Census), utilize spatial statistics techniques (e.g., inverse distance weighting and Kriging) [76, 133] for interpolation, and to visualize derived patterns (e.g., air quality interpolation) in maps.

From the Babylonian world map circa 600 B.C. [67] through the pioneering geographical works of Eratosthenes, the father of geography, to the development of the Mercator projection [146] for sailing and global navigation, and the mapping of rail transport networks [145], cartography and geo-visualization have been pivotal in human innovation for centuries. They enable exploration and navigation across the globe. The rich history of cartography, with its emphasis on geographical accuracy and detail, combined with modern principles of data visualization [44, 47, 154] and human-computer interaction (HCI) [22, 65, 99, 110], opens new avenues for research. By leveraging cartographic insights for spatial orientation and accuracy, alongside data visualization for abstract data representation, and HCI for intuitive user engagement, we have the means to create more comprehensive and accessible visualizations to explore the human body in our Atlas-EHR vision. Figure 2(a) and 2(b) shows two simple illustrations. Figure 2(a) depicts the human circulatory system, mapping veins and arteries critical for blood transport—a system akin to transit stops, essential for oxygen and nutrient delivery and waste removal. 2(b), is a sankey map of the circulatory system, depicting the flow of oxygenated blood (similar to Minard’s flow map of Napoleon’s invasion of Russia [50]) showcasing the integration of statistical graphics into a human anatomy map. More details on challenges and opportunities will be discussed in Section 4.

3.2 Spatial Data at the Scale of Inner Space

Emerging spatial omics technologies are providing researchers with biomedical data at the cellular and molecular level. These technologies analyze RNA molecules within cells. RNA acts as a messenger that takes the genetic instructions from DNA and uses them to produce proteins that the cell needs. The types and amounts of proteins created depend on the cell’s condition. For example, a healthy cell will produce different proteins compared to a diseased cell. By studying RNA, spatial

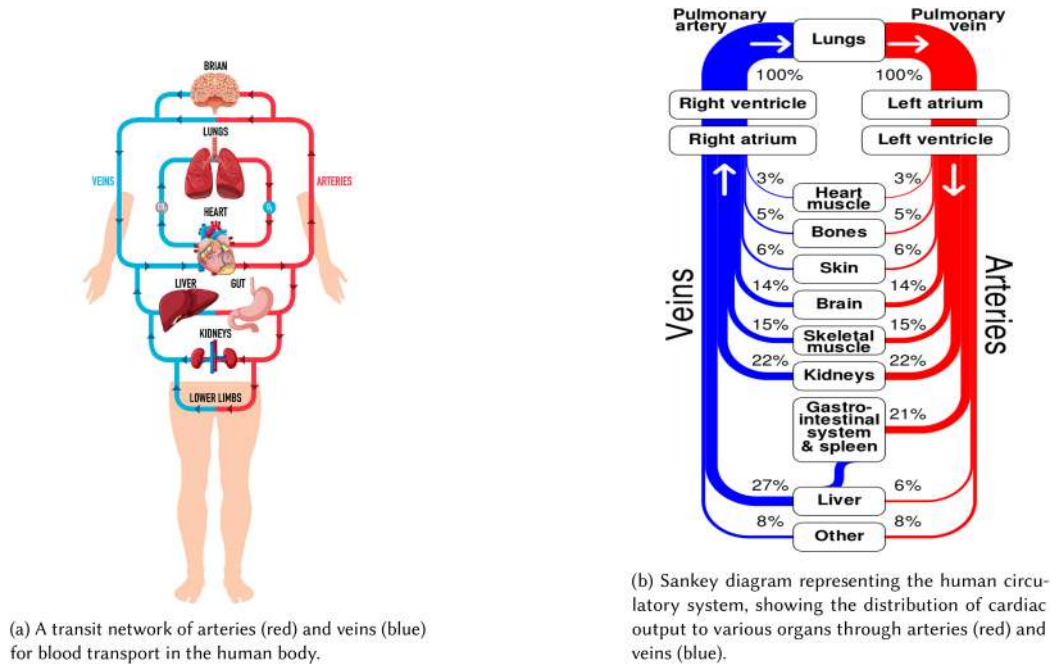


Fig. 2. Spatial maps of the human circulatory system, illustrating the flow and distribution of blood to various organs.

omics reveal which proteins a cell is making under different physiological or pathological states. This gives key insights into cellular and tissue function that are essential for health and survival.

The HCA [129] is a global collaborative effort aimed at creating a detailed map of every human cell type to fundamentally enhance our understanding of health and disease mechanisms. Initiated by researchers around the world, the HCA seeks to characterize cells not only genetically and functionally but also in terms of their spatial distribution within tissues. This ambitious project endeavors to transform our approach to diagnosing and treating diseases by providing an unprecedented level of insight into the cellular basis of human biology. Through its comprehensive cellular mapping, the HCA promises to lay the groundwork for significant advancements in medical research and healthcare and offers a more nuanced understanding of the human body's inner workings.

The **Human Tumor Atlas Network (HTAN)** [68, 128, 152, 161] focuses on developing comprehensive atlases of tumor development to understand cancer evolution through the integration of spatial and temporal data. By mapping the progression of various cancers in detail, HTAN enhances our understanding of cancer biology, which in turn facilitates the creation of more effective therapies. This network represents a concerted effort to dissect the complexities of cancer, using advanced technologies to capture the dynamic changes in tumors over time and in different environments, ultimately contributing to improved cancer diagnosis, treatment and management strategies.

The HCA and the HTAN represent significant strides in biomedical research, but they primarily focus on collecting data and providing tools for basic cell interactions. Studies of more intricate spatial cellular interactions and variations in individual cell composition remain largely unexplored and present significant opportunities for further development within spatial computing research. Moreover, the HuBMAP [83] aims to develop detailed maps of human biomolecules at the

cellular level. A notable achievement of HuBMAP is the creation of Organ Mapping Antibody Panels (OMAPs) for complex spatial imaging in seven human organs. The program also provides computational tools like STELLAR for annotating single-cell data and segmenting specific structures within tissues. The Cellular Senescence Network (SenNet) Program, initiated by the Common Fund, aims to conduct a thorough investigation and differentiation of senescent cells within the human body, across various health conditions, and throughout the human lifespan. This endeavor seeks to map and understand the complex nature of cellular senescence in relation to human health and aging. SenNet leverages advances in single cell analysis, similar to those in HuBMAP, but with a specific emphasis on senescent cells, providing tools, technologies, and a publicly accessible atlas of these cells and their secretions. However, oncologists are interested in understanding the spatial interactions among immune and cancer cells in situ within the **tumor microenvironment (TME)** in order to gain insights into the mechanism of immune therapies and inform the development of new and more effective treatments.

Developing research in this field highlights the need for an automated process to analyze the complex spatial relationships among different cellular subsets and functional states within the TME in the context of **immune checkpoint inhibitor (ICI)** therapy [64, 105]. Furthermore, it is clinically crucial to examine the importance of cell species along with their functional status in a spatially informed manner, due to the clinical implications of interactions in close spatial proximity. Analyzing such spatial interactions helps generate new hypotheses toward discovering disease therapeutics (e.g., immunotherapies for cancer treatment) and could be used in applications such as medical pathology, biomedical research, and microbial ecology. This gives rise to multiple challenges, such as multi-category point patterns, which are heterogeneous and form complicated structural and higher-order spatial interactions. In addition, this can also result in an exponential number of category subsets, which may vary in the strength of their spatial interactions. The proposed Atlas-EHR envisions handling such spatial heterogeneity for both global and local frames of reference and further provides GPU capability to speed up the analysis of exponential multi-categorical relationships while preserving spatial interpretability for oncologists.

4 Accomplishments and Open Research Opportunities in Inner Space

In this section, we discuss the challenges and open research questions related to spatial computing in the human body. The Atlas-EHR processes are related to five broad areas of spatial computing, including sensing, SDMS, spatial pattern mining, positioning, location-based services and navigation, and cartography and visualization. These areas can be described through the completed data life cycle report [12] illustrated in Figure 3 in the context of Atlas-EHR.

The Atlas-EHR data life cycle begins with the “Acquire” phase, where biomedical data is collected using advanced sensing technologies. This includes patient health information from medical imaging, lab results, and wearable devices. The data is then transitioned into the “Clean” phase, stored in SDMS to ensure data integrity, security, and accessibility. These systems play a crucial role in organizing, filtering, and retrieving patient information efficiently. Following data cleansing, the “Use/Reuse” phase involves applying Spatial Pattern Mining techniques to analyze the data, identifying significant patterns and relationships that aid clinical decision-making. This is essential for understanding disease progression and patient responses to treatments, which include steps such as analyze, mine, and model. The next step is the “Publish” phase, where insights are shared and visualized using cartography and visualization tools to transform complex datasets into understandable and actionable formats. This includes workflows for sharing and disseminating data and results. Based on data interpretation and results, as well as feedback from domain experts, the process could iterate back to the data collection stage for additional data and modeling to refine and enhance the system’s accuracy and effectiveness, represented in the “Preserve/Destroy” phase.

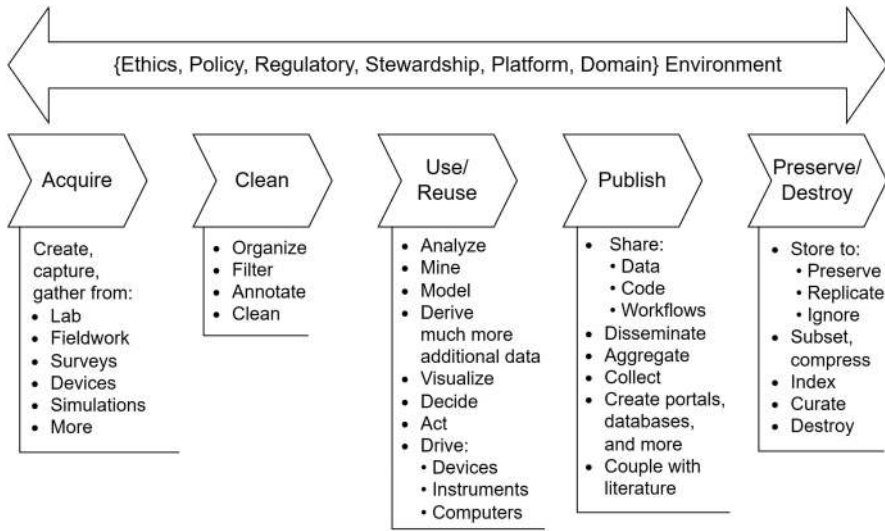


Fig. 3. Overview of Atlas-EHR Across the Data life cycle.

These actions ensure that data remains usable and accessible over time and that it is stored in a manner that supports quick retrieval and compliance with data retention policies.

The unified vision of Atlas-EHR is designed to reduce the time healthcare professionals need to navigate EHR data by integrating advanced sensing, spatial data management, pattern mining, and visualization. This comprehensive, integrated system transforms how medical histories are accessed and interpreted. It is important to note that the workflow depicted may not be strictly sequential or disjoint, as functions of SDMS could overlap across multiple data life cycle phases, such as accessible portals and databases within “Publish”, or subset compression and indexing within “Preserve/Destroy”.

This section identifies challenges and examples within each sub-area, including the development of new spatial data models to accurately represent the intricate spatial relationships within the human body, addressing privacy and ethical concerns during data collection and processing, generating and analyzing large-scale cellular maps, creating spatially-explainable models, managing spatial and patient variability, mitigating spatial bias to ensure fairness, and developing spatially-enabled generative AI techniques for visualizing cellular maps and simulating biological behaviors.

4.1 Sensing

Recent advances in artificial intelligence, machine learning, sensor miniaturization, and battery life have revolutionized wearable smart devices. These devices have become more comfortable, less intrusive, and capable of monitoring a wider range of health metrics [131, 134]. They enable continuous and real-time data collection that is driven by improvements in the Internet of Things (IoT) [2, 87, 120, 127]. For example, devices that monitor and rehabilitate gait in people with Parkinson’s disease (PD) exemplify the practical applications of these technologies. These smart devices, ranging from smartwatches to body-worn sensors, are designed to provide objective, quantitative data on the motor symptoms of PD, such as slowness of movement and involuntary muscle movements [81, 108]. Another innovative development is smart clothing, which excels particularly in cardiovascular risk assessment and monitoring by leveraging sensors like electrocardiograms (ECGs) and photoplethysmography (PPGs). These garments provide continuous measurement of

vital health indicators such as heart rate variability and changes in microvascular blood volume, which are crucial to predict cardiovascular risks [9, 109]. However, several challenges still hinder the widespread adoption of wearables in clinical practice, including concerns about device accuracy, patient privacy, and cost, and the difficulty of separating actionable data from noise. In addition, well-designed trials are needed to fully establish their advantages, all of which are detailed in recent studies [9, 119, 147, 149].

Beyond IoT-enabled wearable and smart devices, medical imaging is a vital technique used in healthcare. It falls within the broader definition of remote sensing, where electromagnetic radiation or sound waves are emitted towards the body, and their interactions provide information for visualizing human internal structures and organs. Although these technologies could provide high-throughput spatial information at great details across different modalities (e.g., MRI and CT scans) and scales (e.g., whole-body, organ, cell, molecular, and genetic), they are limited to providing snapshots and cannot continuously monitor dynamic physiological processes (e.g., blood flow and cardiac function) in real-time. In addition, current medical diagnostics involve the extraction of a sampling tissue, which is difficult to do in sensitive areas, such as the brain. Can we generate some minimally invasive methods for such sensitive areas which can provide more value in early-stage diagnosis?

Towards a Hybrid Future: Merging Wearables and Imaging: While smart wearables provide continuous health data, they have accuracy limitations. Medical imaging gives high-resolution spatial snapshots but lacks dynamic physiological information. Could a hybrid approach that combines sparse wearable data streams with biophysics-based modeling reconstruct a 4D volumetric processes? Examples include real-time cardiovascular dynamics monitoring that requires detailed 4D models of blood flow and cardiac dynamics. Is it possible to capture this through integrating sparse, noisy data from wearable sensors with spatially and temporally coherent biophysical models? Accounting for missing data, modality-specific noise and artifacts, aligning different coordinate systems, fusing data collected at different frequencies, and privacy-preserving analytics provide challenges for current spatial sensing data pipelines. Developing biocompatible sensors that can provide high spatiotemporal resolution signals from sensitive organs without negatively impacting surrounding organs and nervous system pose additional complexities.

AR can improve surgical precision, speed, and safety by providing live visual overlays like critical anatomy in the surgeon's field of view. This approach can reduce complications and accelerate the learning curve for surgeons. Challenges include ensuring accurate tracking and registration of overlays, robust occlusion handling, developing prototype systems that integrate smoothly into surgical workflows, user-centric spatial interaction techniques, and conducting user studies to quantify their value.

4.2 Spatial Database Management systems

The fundamental inquiry focuses on the interactions among spatial entities (e.g., vital organs, their components, subcomponents, tissues, and cells) within the inner space, which depends on an underlying spatial framework defining their relationships. To make this concept more relatable, imagine the human body as a city, where organs represent various departments; for instance, the heart functions as the water pump station, and the lungs act as the air filtration system. These "departments" interact and communicate to maintain the health and efficiency of the city or body. Take the gut-brain axis [21] as an example: it serves as the communication line between the city's food supply chain (gut) and the city management center (brain), which controls the city's mood (emotions) and operational efficiency (digestion and nutrient absorption). Similarly, the heart acts as the central pump in the water system and the lungs as a purification plant, where air entering the lungs is similar to the water that is being filtered. The heart then pumps this "cleaned" air

(oxygen-rich blood) through the body's "pipeline system" to ensure every part receives the "fresh-water" necessary for proper function. SDMS act as the city's mapping and information hub, organizing data about the interactions between departments and aiding in the understanding and optimization of the body's functions.

Spatial Data Models: While OGC's simple features (e.g., multipoints, multilines, and multipolygons) [48] offer a solid foundation for representing common shapes on a map through a spatial reference system, accurately representing and analyzing spatial relationships in inner space can be challenging. These include the nonrigid and flexible nature of soft tissues and organs, their dynamic changes, ability to compress and expand, pathological changes that alter spatial boundaries, the existence of multiscale structures, the distortion from projecting three-dimensional organs onto two-dimensional maps, and the lack of a spatial reference system tailored to the human body. Addressing these issues requires advancing beyond traditional data models and operations designed for geographic spatial entities.

To illustrate, topological relationships (e.g., disjoint, meet, and overlap) highlight properties that remain constant under continuous transformations, whereas nontopological relationships (e.g., metric relationships such as distance and angle) can change with transformations. These are typically defined through the existing 9-Intersection Model and the queen-based neighborhood directional model. However, capturing the intricate spatial relationships within the human body requires considering a combination of these relationships, extending beyond pairs to include non-distance-based spatial associations (e.g., *surrounded by*) that are invariant to elastic deformation.

For instance, understanding the relationship between the heart and lungs in the cardiovascular and respiratory systems involves recognizing how oxygen-poor blood is pumped from the heart to the lungs (a topological relationship), where it undergoes oxygenation (a nontopological process involving gas exchange), and is then returned to the heart for systemic circulation. Therefore, spatial data models must incorporate anatomical pathways (veins and arteries) and physiological processes (oxygenation levels, blood flow rates) to examine a complex interplay between structure (topology) and function (nontopology) across vital organs. Similarly, the interaction between the digestive and circulatory systems involves the liver's role in processing nutrients from the intestine and its complex vascular connections to other organs. Blood from the digestive tract is transported to the liver via the portal vein (topological), where it is filtered and metabolized (nontopological, involving chemical processes). The liver then releases nutrients into the bloodstream for distribution throughout the body, exemplifying the integration of spatial pathways with biochemical transformations across different organs.

Big Spatial Data at Inner Space Scale: Current studies [13, 16, 86, 153, 169] in spatial omics and cellular mapping often analyze a limited number of pathology-driven fields of view (FOVs), typically ranging from five to ten, to investigate spatial cellular interactions within the TME (i.e., one slide). However, comprehensively understanding the TME requires analyzing hundreds of FOVs, potentially reaching 300 for a detailed picture. If the aim is to capture all tissue regions (normal, boundary, and malignant), the number of FOVs can exceed 1300. A recent study [13] has captured entire slide images, generating massive datasets reaching five terabytes. Considering that a single tumor biopsy is typically divided into several slides for targeted analysis, mapping an entire biopsy at high resolution could require up to 100 slides. If the tumor affects multiple organs and involves numerous patients with similar conditions, the data volume could easily scale to hundreds of petabytes.

This big spatial omics data is essential because the spatial relationships and patterns within it are crucial for understanding biological processes and cannot be fully captured without comprehensive and integrated analysis methods. Although parallel processing and time-staggered analysis can distribute the computational load, they do not address the fundamental challenges of

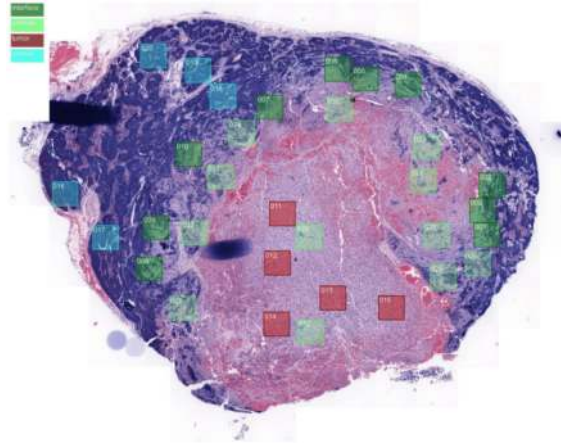


Fig. 4. A virtual H&E tissue slide including pathology-driven fields of view namely, infiltrate, interface, tumor, and normal [105].

data storage, retrieval, and real-time analysis demands. Distributed systems and parallel processing across different machines or sequential analysis over time can mitigate some computational burdens, but they do not directly solve the challenge of integrating diverse data types (e.g., imaging, genomic, and proteomic) to derive meaningful insights. Additional complexities arise due to variability in biological processing across different organs (i.e., different locations within the body) and their associated subcomponents, which can vary significantly and pose significant challenges for existing spatial big data management systems. For instance, diseases such as Alzheimer's or Parkinson's necessitate time-critical analyses, requiring real-time processing of spatial data to monitor physiological changes in the brain over time.

This demonstrates the potential and need to design new spatial data models and modify existing spatial indexing, query processing, and optimization to efficiently handle massive spatial data on a scale. For example, is it necessary to capture the whole slide image? To reduce the expense of storing hundreds of petabytes of data, would it be possible to consider a range query or a nearest neighbor algorithm to estimate the spatial distribution of the whole slide image, perhaps followed by imposing a non-uniform spatial grid index to sample FOVs? With an estimate of the spatial distribution of the whole slide image, is it possible to define the cost and route over a spatial network of FOVs to reach areas associated with the high proliferation of tumor cells?

Figure 4 shows pathology-driven FOVs from a virtual H&E tissue slide, which includes tumor, infiltrate, interface, and normal FOVs. The current inspection process is expensive and time-consuming due to the need for domain experts (e.g., pathologists) to manually analyze tissue slides and label them according to response to antibody staining. The current solution is to capture the whole slide image by imposing an overlapping grid structure; however, the associated data storage cost is high. We envision an algorithm that begins with a stochastic approach to generate random FOVs over the whole slide image. This is followed by a spatial network, where nodes are individual FOVs and edges describe the similarity between FOVs based on a summary statistic to estimate the spatial distribution at the FOV-level. This process can be repeated until an optimal spatial distribution is reached, meaning when adequate FOV group samples are generated with high intra-similarity and minimum inter-similarity between groups. However, additional complexities may arise due to the impact of the edge effect and signal-to-noise ratio in estimating spatial distribution.

Spatial Physical Design for Omics ML: The following research questions highlight the lack of spatial components in recent machine learning algorithms, including multitask and federated learning. Is it possible to model the underlying spatial distribution of sampled FOVs from specific tissues (e.g., skin, kidney, and lungs) to define spatial indices over regions that share similar spatial configurations? If successful, this may result in millions to billions of parameters of pre-trained models related to specific tasks on big spatial omics data. What is the best way to define a standard spatial projection and reference system to leverage large data samples with different formats across multiple sources? Do pre-trained models need to be stored on disk or memory, or do we need to define intermediate stages on the database that save some parameters that can then be fine-tuned for other specific tasks? How can spatial omics data be most effectively prepared at scale to be used by machine learning models on related downstream tasks?

Multitask learning aims to improve the performance of individual tasks, where a model is trained on multiple tasks simultaneously by leveraging the shared representations learned across all of them. Federated learning has also gained attention in recent years. Federated learning is a distributed ML technique where a model is trained across multiple decentralized devices without sharing the raw data with a central server. This becomes important since patient privacy and the propriety nature of medical data prevent many organizations (e.g., hospitals and medical research establishments) from publishing their datasets. Federated learning is an effort to address this issue by only transferring model parameters while training the model on locally stored data. However, such algorithms may not be generalizable in all domains, since recent works [86, 117] have highlighted that analyzing spatial omics data is location-specific within the body (e.g., brain, kidney, and lungs), and separate training models are needed to better understand the structure, composition, and underlying spatial patterns. This demonstrates the critical role of spatial database management and relevant techniques (e.g., spatial indexing) for preparing these data at scale to be used by machine learning models on related downstream tasks (e.g., classification and segmentation).

Spatial Query and Modeling for Blood Flow: Understanding and modeling blood flow challenges traditional spatial data models and querying techniques due to the dynamic nature of the circulatory system. Blood vessels pulsate and change shape and size with each heartbeat. Current spatial query techniques (e.g., R-trees, Quad-trees, and Voronoi diagrams) rely on static geographic data and fail to account for the continuous changes in artery and vein diameters that significantly affect flow and pressure. Also, standard spatial queries fail to address the unidirectional flow in the arteries away from the heart and in the veins to the heart. Moreover, these models do not handle the complex interplay between different vessel types at varying scales, from macroscopic to microscopic.

As an example, cardiologist optimizing a catheter insertion pathway in the coronary arteries to ensure minimal disruption to blood flow and avoidance of critical areas. The baseline spatial SQL query used for this procedure might look as follows:

```
SELECT vessel_id, ST_AsText(ST_GeomFromText(vessel_geometry)) AS vessel_geometry, blood_flow_velocity,
       pressure
FROM blood_vessels
WHERE vessel_type = 'artery'
      AND pressure < 120
ORDER BY blood_flow_velocity DESC;
```

Fig. 5. Baseline query to select high-velocity arteries with low pressure.

This baseline query (refer to Figure 5) assumes static vessel geometries and fails to reflect the dynamic and time-dependent changes in vessel shape, size, blood flow velocity, and pressure. It

```

WITH DynamicVessels AS (
  SELECT
    vessel_id,
    ST_MakeLine(ST_Point(x, y, timestamp), ST_Point(x2, y2, timestamp2)) AS vessel_geometry,
    dynamic_flow_rate(blood_flow_velocity, heart_rate, timestamp) AS flow_rate,
    dynamic_pressure(pressure, elasticity, timestamp) AS adjusted_pressure,
    timestamp
  FROM real_time_blood_vessel_data
  WHERE vessel_type = 'artery'
    AND dynamic_flow_rate(blood_flow_velocity, heart_rate, timestamp) > 0
    AND dynamic_pressure(pressure, elasticity, timestamp) < 120
)

SELECT vessel_id, ST_AsText(vessel_geometry), flow_rate, adjusted_pressure, timestamp
FROM DynamicVessels
ORDER BY flow_rate DESC, timestamp ASC;

```

Fig. 6. An illustrative example of optimized query for dynamic modeling of blood flow.

also does not adapt to the unidirectional flow or the elastic deformation of blood vessels. Finally, it is limited by its inability to handle complex spatial interactions at different scales.

To address these challenges, one may consider the following optimized version, as shown in Figure 6. This code snippet is a conceptual query designed to serve as an illustrative example, highlighting several novel features that address the drawbacks of existing methodologies. However, the exploration of these improvements requires further investigation for dynamic modeling of blood flow in the future. The dynamic functions include the rate and pressure of blood flow, while the static functions encompass real-time heart rate, vessel elasticity, and blood flow over time. This approach represents dynamic modeling of the sustained and pulsatile flow of blood, adjusting to relevant physiological conditions over time.

Besides, time can be explicitly factored into spatial data using the *ST_Point(x, y, timestamp)* function; temporal data can be embedded into the calculations to consider the variables that develop in time. Such a specified integration of time ensures that both present and historical physiological states are included in the query, yielding timely and reliable information relevant to real-time decision-making. It passes through lines with acceptable flow rates in the *dynamic_flow_rate* function, working through temporal data, which allows a person to avoid physiologically unacceptable paths brought about by traditional spatial queries. In addition, it gets adjusted to elastic deformation of blood vessels by real-time sensor and temporal changes to preserve spatial relationships and flow characteristics adapted to physiological variations.

Like the baseline query, this function supports using complex spatial relationships with advanced functions, such as *ST_MakeLine* and *ST_AsText*. It provides multiscale modeling with explicit temporal features, where this holistic approach captures interactions among various types of vessels at different scales and gives a detailed insight into the time-evolving vascular system.

An objective performance measure of this process is to minimize the **Query Response Time (QRT)** while maintaining a high **query accuracy (QA)** for spatial queries involving blood flow challenges, such as unidirectional flow and dynamic changes over time. The efficiency metric shall be defined as $\text{Efficiency Metric} = \frac{\text{QRT}}{\text{QA}}$ to balance the need for accurate and timely information. Ideally, QA tends to be 100%, QRT is minimized, and the efficiency metric should tend toward a high value. Theoretically, the latter is high, but depends on practical constraints. The processing

and modeling of queries may face additional complexities and challenges due to patient conditions such as hemorrhage (e.g., internal bleeding), external factors (e.g., open skin wound and bleeding), or rare events (e.g., stroke). These factors need to be taken into account when addressing this underlying problem and when considering the limitations of existing spatial data models and query processing.

4.3 Spatial Pattern Mining

Spatially-Explainable Prediction Models: Domain scientists (e.g., oncologists and pathologists) are interested in understanding why patients respond differently to a candidate immunotherapies. Success hinges on complex spatial interactions between cancer and immune cells within TMEs. Studying these manually is increasingly challenging due to the exploding volume and richness of spatial omics and cellular maps generated by modern imaging. For example, a single tissue sample contains millions of cells representing over 60 distinct cell sub-types, which results in a potential million trillion spatial cell interactions to investigate. In addition, the strength of these interactions can vary, with some interactions being more biologically relevant than others, requiring the model to represent these distinct levels of interaction.

A possible research direction is directly using advanced classification models (e.g., adaboost, xgboost, KNN, and SVM) and state-of-the-art DNNs for downstream tasks (e.g., predictions and classifications). However, these models do not provide spatial explainability. Hence, is it possible to describe the clinical outcomes of an intervention using spatial concepts (e.g., *touch*, *inside*, and *surrounded by*) and patterns (e.g., colocations, spatial outlier, and hotspot-colocations)? The early spatial approaches used hand-crafted measures from classical spatial statistics (e.g., Ripley's cross-k function and G-cross) [8, 37] and spatial data mining (e.g., participation index) [69, 89, 141, 166], are primarily based on distance relationships. These do not model directional spatial relationships such as *surrounded by*, which may be of biological significance.

Explainable artificial intelligence (XAI) [6, 26, 33] encompasses algorithms designed to enhance transparency and trust by providing understandable explanations for their decisions while maintaining acceptable performance. This becomes even more crucial for spatially-explainable AI approaches when analyzing spatial patterns in multi-category point sets like spatial omics and cellular maps. This research is of significant importance in the advancement of ICI cancer therapies, where the effectiveness hinges on spatial proximity between immune cells and cancer cells, triggering crucial biochemical interactions. Our recent work [41, 88] has shown promise for machine-constructed spatial cell interactions in modeling graph-based spatial relationships, outperforming state-of-the-art methods in separating responder from non-responder classes. A novel neural network architecture was designed for the cellular map data to discover any significant spatially-explainable patterns that could not be captured by traditional spatial data mining or state-of-the-art DNN techniques. This framework introduces two new layers to the neural network to classify cell interactions. The first layer models cellular maps belonging to different distributions (e.g., clustering versus even distribution), and the second distinguishes between cell-cell instances that belong to different types by learning the importance of each distinctive pair.

Our preliminary work [41] sought to identify the spatial differences that distinguish responders and non-responders at the core of the tumor. Figure 7(a) shows the multi-category point sets for all cell species in a responder class compared to a non-responder class. The significance of the identified cell interactions, that is, the machine-constructed spatial cell interactions for different category subsets, was evaluated according to the importance of feature permutation. The most important characteristic was identified as spatial cell interactions among <tumor, macrophage, and neutrophils>, where the macrophage is a type of immune cell, and the neutrophil is a type of immune cell. This analysis revealed (Figure 7(b)) that macrophages were associated with tumor

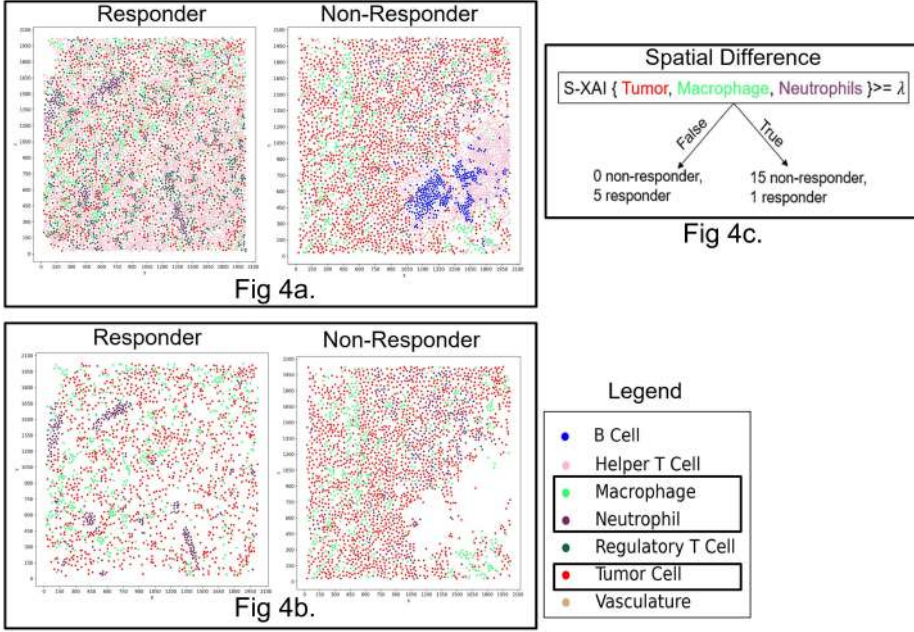


Fig. 7. (a) Multi-category point sets of the responder and non-responder (b) Point patterns detailing spatial arrangements between Tumor cells, Neutrophils, and Macrophages. (c) A spatially-explainable decision tree to distinguish between responder and non-responder at the tumor core.

cells in both the responder and non-responder classes. It also showed that neutrophils formed small sub-clusters in the responder tissue sample, while in the non-responder tissue sample, they surrounded the tumor-associated macrophage. A spatially-explainable AI classifier (Figure 7(c)) uses the spatial differences between these cells as explanatory features to distinguish between responder and non-responder multi-category point sets.

However, this is just the beginning. There are many other opportunities to explore, including the development of AI methods that can explain spatial patterns, such as spatial hotspots and anomalies. We should also consider the possibility of detecting even more complex spatial interactions using explainable data mining, not only in cellular maps in oncology, but also in maps from other domains. Additionally, we need to address challenges related to both topological (e.g., disjoint and overlap) and non-topological relationships (e.g., angle, distance, area, and directions) within a diverse spatial graph containing millions of cells. It is important to consider the temporal dynamics of cells as well. Cells may grow and divide, enter a resting phase, become activated in response to a signal (e.g., infectious or injury) or metabolic changes, or stop growing due to age or damage. All these factors could create complex hyper relationships within a model or diffusion of models, and it is crucial to preserve spatial explanations while considering this additional complexity. These are the open questions that need to be addressed for spatially-explainable prediction models.

Spatial Variability Aware AI Models: Spatial variability introduces additional complexities to the analysis of multi-category point sets. Spatial variability refers to the inherent heterogeneity and variations observed in spatial patterns, structures, or arrangements across different locations. For example, in Figure 8, the three selected multi-category point sets (from Figure 4) depicting place-types (e.g., tumor-core, tumor-margin, and normal regions) illustrate significant spatial variability, revealing variations in cell population across different locations within a single tissue sample. Beyond variability across place-types, intricacies emerge within their sub-categories. For instance,

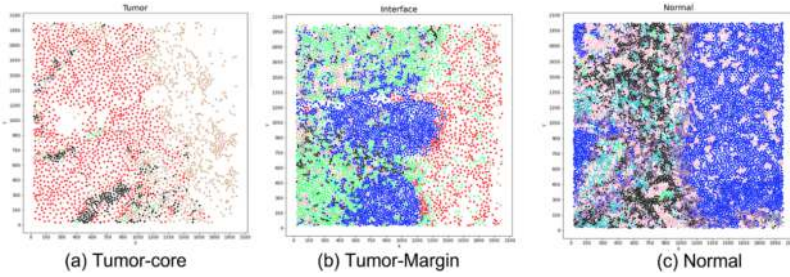


Fig. 8. Three pathology-driven fields of view are categorized based on tumor infiltration.

tumor margins can be classified as intact or infiltrated based on tumor boundary properties. Intact regions, displaying negative response, have boundaries between immune and tumor cells remaining intact. Infiltrated regions, on the other hand, can represent either a favorable response (immune cells invading and destroying cancer) or a negative response (tumor cells attacking immune cells). Spatial variability can further extend to sub-regions within a place-type, where specific patterns like hotspots might be more prevalent than others, such as colocations and anomalies.

While recent studies [111, 168], offers insight into TME heterogeneity, modeling its spatial variability is challenging due to factors such as rapid cancer cell proliferation, genetic instability, and the presence of unknown mediator cells with immune and target cells. These complexities demonstrate the need for spatial variability aware pattern mining in analyzing cellular maps in inner space.

Geographically Weighted Regression (GWR) [17] is a traditional non-parametric model that learns location-specific maps but struggles with complex predictions. Recent work such as SVANN [59] and [165] study spatial variability through deep learning considering location-dependent weights and learning space partitionings of heterogeneous data using weight-sharing mechanisms. However, techniques require dense training data or have limited ability to handle significant spatial variability within a single place-type. Moreover, they are designed for geographic data in Euclidean space and miss the complexities of spatial variability in non-Euclidean spaces like point clouds and multi-category point sets. The advancement of CNNs [23, 62, 84] in pattern recognition has inspired their adaptation to analyze 2D / 3D point cloud data without the need for expensive conversion steps. Pioneering DNNs like PointNet [122], DGCNN [157], and the more recent PointTransformers [171] have significantly improved the processing of point cloud data for pattern recognition. However, these methods primarily focus on numerical attributes, which limits their ability to capture the complex spatial relationships among various categories of points. SAMCNet [41] addresses this gap by emphasizing spatial interactions across diverse point types but does not adequately account for *spatial variability*, with a preference for the scalar over the weights of the map.

Our recent work [42] explores a spatial ensemble framework, learning unique decision functions for different place types (e.g., tumor-core and tumor-margin) and hypothesizes that a spatial ensemble of base classifiers would enhance prediction strength and model interpretability. The network parameter is a map varying across place-types rather than a scalar as used in traditional size-fits-all approaches. Additionally, this work investigates more flexible spatial domain adaptation layers to tackle issues with insufficient learning samples, using a weight-sharing mechanism for location-independent and dependent layers. Location-independent layers are learned using all cellular maps across all regions, while subsequent layers are fine-tuned to capture location-dependent features for the target domain classifier. This approach mirrors the innovation seen in foundation models [15], where it is envisioned that location-independent features (domain-independent) in a target domain classifier can be derived from the original foundation model without the need for

task-specific data from scratch. However, location-specific features that differ across place-types still necessitate fine-tuning. Recent advances include the use of generative adversarial networks [30] for domain adaptation [155, 172] that leverage minmax optimization to train domain-specific discriminators and a generative model that seeks to produce domain-invariant features. The application of these techniques to spatial datasets has yet to be fully explored [100, 101].

There are other emerging techniques such as knowledge-guided methods that customize components of a neural network using physical properties like loss function (e.g., regularizer), initialization (e.g., synthetic data), and architecture to represent relationships among physical variables. This area of research is known as physics-guided and knowledge-guided machine learning [75, 77]. However, beyond these physical properties, can we incorporate techniques from chemical and biological processes that occur within the human body into DNN techniques?

4.4 Positioning, Navigation, and Location-based Services

Computer-Vision Spatial Positioning: Computer-vision positioning [150] is a technique that aims to determine the position and orientation of objects by analyzing visual data, such as images. This technique also can provide coordinates and descriptive text corresponding to the scene being analyzed. However, directly applying these techniques to inner space may not work correctly due to patient variability (i.e., patient-specific) and the non-rigid structure of human organs, muscles, blood vessels, and so on. For instance, the size and position of the liver can vary significantly due to factors like age, gender, and the presence of certain diseases. Patient variability could be classified based on a hierarchy of demographic, health, and physical fitness information. From a high-level view, patients may be classified by their age, race, ethnicity, and gender, factors that have a high impact on the inner space of human physiology. Additional complexities may arise due to the patient's medical history (e.g., past injuries, chronic illness, etc.) and physical fitness. At this level, computer vision positioning faces a great challenge due to the heterogeneity of the data at multiple phases. Even within a single individual, organs and their sub-components can exhibit significant variability in morphology. This spatial variability adds another layer of complexity for positioning and analysis.

A possible solution may be to define a heterogeneous data-aware hierarchy to split the data to account for patient variability. Such a hierarchy could be specified as a procedure to characterize a social/fair/mathematical norm such that in-depth features (e.g., detailed physical or health history) impact how the dataset is partitioned. Beyond patient variability, the non-rigid structure of human organs could highly impact how computer-vision-positioning technology should be modified and adapted to inner space. Human organs and muscles, including the heart, kidneys, liver, and cardiac and smooth muscles, can change shape and size in response to various physiological and pathological processes, and the frequency of these changes can vary widely depending on the specific organ/muscle and the conditions it is subjected to. For example, the heart expands, and contracts with each beat, and the size of the ventricles can change in response to changes in fluid volume. The lungs can expand and contract with each breath, and the size of the air spaces can change in breathing rate or lung volume. This elastic deformation of human organs with high temporal variability calls into question the generalizability of computer vision-positioning. Is it possible to train a learning algorithm on images of a certain physical structure of some organs and use those images to determine the elastic deformation of the same organ at the next temporal step?

Navigation and Recommending Paths in Innerspace: The current health systems have not fully studied the capability to calculate and suggest optimal paths for navigating medical interventions using real-time spatiotemporal data. This is mainly an optimization problem wherein one has to decide on the optimal route for delivering treatments or performing surgical procedures within the human body. The optimality for these paths is defined by minimizing

risk, maximizing efficiency, and improving patient outcomes. However, the human body lacks standardized spatial reference systems, which makes recommending paths and navigation in Innerspace highly difficult. Other research challenges include crafting a dynamic spatial model of the human body to consistently represent anatomical and physiological changes over time and continuous health data management with robust privacy-preserving and ethical guidelines.

The input for this problem may be characterized by multi-modal patient data. This includes high-resolution medical imaging data derived from MRI and CT scans, continuous real-time data from wearable sensors monitoring vital signals, and periodic examination results like lab tests and clinical measurements. The map must accurately represent critical structures such as bones, major blood vessels, nerves, organs, and tissues to avoid damage during procedures. For example, it is crucial not only to identify a tumor within an organ but also to determine the distance between the tumor and neighboring anatomical structures or the location of a blockage in a blood vessel. Clinical notes and historical health records are integrated into a continuously updated 3D anatomical map and dynamic health model, which reflects both the current and historical health status of the patient.

A suitable measure is required for this framework to evaluate the success of measuring optimal navigation paths. One such recommended metric is the “**Navigation Efficiency Index (NEI)**,” which considers factors like path length, procedural time, avoidance of critical structures, and recovery time of patients. Several constraints need to be taken care of to meet the objectives of this system: The avoidance of major arteries and nerves, minimization of procedural invasiveness, and synchronization of treatment with the patient’s biological rhythms are what medical constraints take into account. A good example is during the preparation for a procedure, where the system should be able to calculate an optimal path for a catheter to reach the plaque while avoiding critical structures such as heart valves and major arteries. Technical constraints entail the integration and processing of heterogeneous data sources in real time, with the ability of the system to scale up when dealing with complex and variable patient data.

One approach to address such challenges is to develop a hierarchical spatial reference framework that captures the anatomical landmarks and structures with increasing levels of granularity. The first step will be to create high-resolution 3D anatomical maps from appropriate medical image data, including MRI and CT scans. These maps should span the range of anatomical structures, from organs, blood vessels, nerves, and bones to the substructures of these components. At the top level, one can think of the major organ systems (e.g., cardiovascular and respiratory); the next level would comprise individual organs (e.g., heart and lungs), and further levels would include substructures within organs (e.g., heart valves and bronchioles). Any given level must have a set of reference points or landmarks that can be used to register information from other sources with the anatomical maps. This also requires fixing the reference points so that there may be consistency in spatial orientation and navigation among different patients.

Human anatomy is unique and varies; therefore, the framework must also accommodate this fact. Consequently, this translates into context-aware indexing because such algorithms take into account the physiological state of the patient, such as changes in the positions and shapes of organs due to breathing, heartbeat, and other physiological processes. The system must continuously aggregate data on the patient from multiple sources: historical records on health, imaging studies, and sensor data in real time. This will give it a comprehensive view of anatomical variations across different patients. Continuous learning mechanisms should be built to enable model updates regularly to incorporate new patient data. For example, neural networks could learn to recognize patterns and resize spatial indices according to the understanding gained from the aggregated data. The learning process will be ongoing, thereby keeping the spatial reference framework accurate and updated. When all of these challenges have been innovatively addressed,

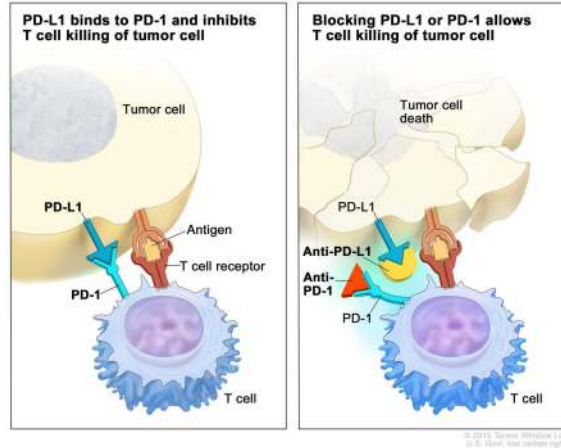


Fig. 9. Importance of spatial analysis in evaluating the effectiveness of ICI therapy in killing tumor cells [71].

with continued enhancements, the Atlas-EHR system calculates and guides efficient paths for medical interventions to increase precision and effectiveness in healthcare delivery.

4.5 Cartography and Visualization

Atlas-EHR, an inner space information system, serves as an integrated platform that includes the previously discussed techniques to tackle the outlined challenges. This unified framework is adept at collecting, managing, analyzing, and visualizing EHR data. Specifically, it incorporates spatially-explainable AI approaches tailored for various common pattern types. These include spatial colocation among different cell subsets, identification of cancerous and metastasized cell hotspots, and mapping the telecommunication pathways from the brain to vital organs. The system can also integrate these patterns, considering the complex nature of EHR data.

A key principle for understanding the significance of problems, common patterns, and relationships in data lies in exploratory analysis and visualization before employing advanced computational techniques, such as machine learning. In cancer research, understanding certain spatial cell interactions requires the selection of contrasting colors, sizes, and other visualization components to emphasize specific spatial patterns. However, existing GIS software has limited capabilities for automating the visualization of cells of different classes. This results in extensive manual effort. Considering the increasing number of identified cell types, their functional characteristics, and the thousands of samples derived from specific tissues, the cost of manual visualization can increase exponentially. The question arises whether we can enhance existing GIS software capabilities through the innovation of generative AI algorithms suitable for inner space exploration. More specifically, is it feasible to implement a spatially-aware generative AI model that enables end-users to easily visualize cellular maps and distinguish specific spatial patterns with minimal commands?

This functionality can be further expanded to generate illustrations representing biological behaviors across various scales in the human body, encompassing interactions between organs and even at the cellular level. For example, Figure 9 shows two diagrams contrasting the importance of immune ICI therapy in two spatial contexts. The left diagram shows a T cell that cannot destroy the cancer cell due to the spatial association between the tumor and the immune cell. The diagram on the right illustrates a killer T cell that can kill cancer cells due to a blockade (such as anti-PD-L1) at the checkpoint gate (between PD-1 and PD-L1). Existing visualization requires training

individuals with a biology background, who may need extensive manual efforts to emphasize the proper biological processes while delivering an easy-to-understand message. Therefore, it is worth investigating whether spatially-aware AI models can represent similar biological behaviors for creating diagrams and examples that are more accessible and interpretable to a broader audience.

5 Discussion: Bridging the Physical and Digital Space with Atlas-EHR

The successful results of research on Atlas-EHR hold enormous potential for various applications. Advanced spatial computing techniques might be used in precision medicine to fight complex diseases (e.g., cancer) by combining genomic data (at a fine biomedical scale) with other EHR data of patients to create customized treatment plans (e.g., targeting particular genetic mutations) and enhance informed decision-making to maximize clinical outcomes for each individual. Spatial computing methods could support real-time monitoring of the body's inner space to manage chronic disease (e.g., diabetes and heart disease) and predict abnormal events (e.g., high blood pressure and blood clots), enabling doctors to respond quickly to changes in patient's conditions. Spatiotemporally organized 4D map-based-simulation could enhance the existing EHR platform for grouping and managing patients' medical history (e.g., lab results, medications, and radiology images) based on spatial information in inner spaces such as the human body organs to facilitate more efficient and effective querying and visualizing of relevant information to improve prognostics and treatment planning. Spatially-aware surgery could provide surgeons more in-depth understanding of patient-specific anatomy and physiology, in which inner space, together with a simulation system, would allow them to plan and visualize complex surgeries with greater precision, reducing the need for invasive procedures and minimizing the risk of surgical complications. These are but a few examples of our vision. In short, Atlas-EHR has the potential to revolutionize the healthcare industry. In the following, we aim to highlight some of the specific examples with a diverse range of medical fields.

Examples Across Diverse Medical Fields: In Orthopedics, Atlas-EHR can provide a detailed 3D models of a patient's anatomy. This could include an in-depth analysis of fractures, deformities, or joint issues from all angles prior to surgery. Surgeons can use these models to precisely plan the location of incisions, implant sizing, bone resections, and so on. Moreover, these maps could be used for patient education by visualizing injuries and describing proposed treatments to help patients better understand their condition. Seeing personalized anatomical models improves patient engagement in shared decision making. Atlas-EHR can leverages motion sensors to track patient movement and progress through post-op rehab exercises. This helps clinicians ensure patients are recovering properly after orthopedic procedures.

Traditional ear-nose and throat (ENT) assessments rely on limited 2D imaging techniques like X-rays and CT scans. While these provide valuable snapshots, they lack the depth perception and real-time interaction crucial for understanding complex ENT structures and their functions. The conversion of 2D scans into interactive 3D models enable doctors to virtually explore intricate ENT structures like the sinuses, inner ear, and vocal cords. This provides a deeper understanding of spatial relationships and potential abnormalities compared to static images. For intricate ENT procedures involving the sinuses, larynx, or inner ear, Atlas-EHR holds immense potential. Imagine surgeons virtually exploring a patient's unique sinus anatomy before surgery, planning implant placement for complex facial reconstruction, or visualizing vocal cord movements in real-time to diagnose voice disorders.

Another application of Atlas-EHR includes cardiology. Individual variations in heart sizes, shapes, and vessel pathways are crucial for successful procedures. Atlas-EHR can create patient-specific 3D models reconstructed from their unique medical data, providing an intimate virtual replica of their cardiovascular system. Atlas-EHR allows surgeons to plan and rehearse coronary

stent insertions within the patient's virtual heart model. This simulated environment enables them to optimize stent size, placement, and approach, minimizing risks and potential complications during the actual procedure.

6 Conclusion and Future Work

In light of the success of applications such as Google Maps and Google Earth to communicate big data (e.g., satellite imagery and census) to a broad audience efficiently, we argue that spatial computing has a similar role to play in the biomedical domain. To this end, we envision Atlas-EHR, a spatially organized EHR data for reducing the time needed by healthcare to understand patient history via the next generation of biomedical decision support. We discuss a few challenges and opportunities for five broad areas of spatial computing research to explore the human body. For each domain, we list some of the recent accomplishments and then highlight research needs to better solve challenging problems in this space.

For the field of spatial computing, which has been pushing the boundaries of what is possible, the next frontier lies within the human body itself. Exploring inner space computationally will require new thinking and new approaches. ERHs contain a wealth of data that is invaluable to biomedical research, but we need to rethink how we store and extract this data to its full potential. Atlas-EHR provides a conceptual roadmap for the future of spatial computing within the healthcare industry, enabling us to store and analyze EHR data using spatial computing principles. By doing so, we can unlock new insights into complex diseases and develop tailored treatment plans that will maximize patient outcomes.

In our future work, we plan to provide a detailed analysis of challenges related to data collection, retrieval, management processes, and ethical considerations. Challenges associated with a complete data life cycle for collecting comprehensive and detailed user data to construct an accurate personal profile. The complexities surrounding the formulation of data retrieval queries, such as how changes in the human body are propagated into the system and affect information retrieval. We also plan to investigate on ethical concerns and the potential for misuse of the system. This may include the possibility of inherent biases within the system, potentially favoring certain demographics over others (e.g., disparities based on body type, gender, or unique physical characteristics), without suggesting corrective measures. The risk of data breaches leads to the denial of insurance claims. Finally, we will extend our investigation to an examination of the capabilities of Atlas EHR in facilitating the integration of intermediary imaging modalities, including light sheet microscopy [113], micro-CT [126], and electron microscopy (EM) [78], thus improving the comprehensiveness and precision of diagnostic processes.

Appendix

A Overview of Existing EHR Platform

Major EHR systems, including Epic Systems, contain extensive patient histories that span years or even decades of medical data. These detailed records present healthcare professionals with the challenge of efficiently navigating massive amounts of complex information (see Figure 10 for an example EHR interface). The current temporal format of EHRs makes it difficult for medical staff to uncover and analyze the intricate connections between diverse types of patient data gathered by various sensing technologies across multiple scales and time periods. For instance, physicians may struggle to identify correlations between whole-body scan results from years ago and cellular-level test results from a recent visit. The extensive breadth and depth of EHRs, which contain measurements ranging from macro-level bodily systems down to micro-level cellular components, compounds this difficulty.

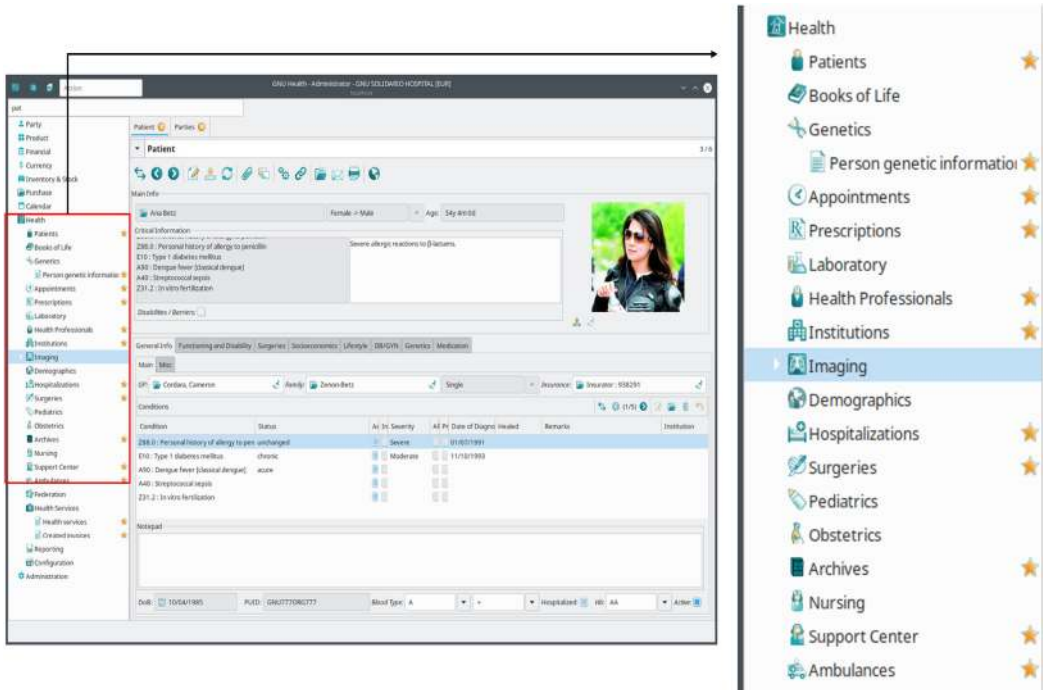


Fig. 10. An existing EHR software without a spatially-defined taxonomy.

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